

Analytics for a Better World

Predicting while optimizing: mobile clinic routing with learned constraints

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joint work with Mayukh Ghosh and Chintan Amrit

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Chapter 6 of Mayukh Ghosh's PhD thesis, *Domain Driven Analytics*



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A forecast can change when the plan changes

- A mobile clinic chooses which wards to serve and how to travel between them.
- Visits nearby may reduce demand through cannibalisation or increase it through mobilisation.
- A demand forecast fixed before routing can miss this dependence.

Today's question

What changes when the same predictor becomes a constraint inside the routing model?

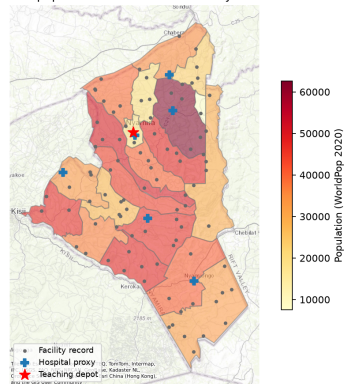
A synthetic teaching companion to Mayukh Ghosh's Chapter 6, with Chintan Amrit and Joaquim Gromicho. The outcomes shown here are simulated.

One county, twenty candidate service points

- Nyamira County, Kenya.
- Ward centroids snapped to an OSM drive network.
- An open path from Township, with at most eight serviced wards.
- The teaching depot also receives service and counts as a stop.

Actual clinic sites and the referral hospital's coordinates would be needed for operational use.

Nyamira: population and historical facility locations



Boundaries: GADM 4.1; population: WorldPop; facilities: Maina et al. 2019

Data sources and what they establish

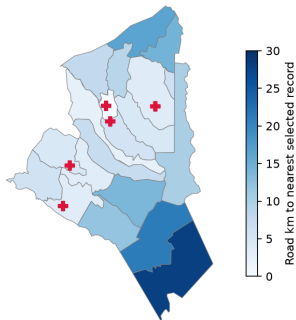
Boundaries	GADM 4.1. Downloaded for local academic use; boundary files are not redistributed.
Population	WorldPop 2020, 1 km, UN-adjusted.
Roads	OpenStreetMap contributors; directed shortest-path distances through Pandana.
Facilities	Maina et al. (2019), 98 historical records; six hospital labels used as an accessibility proxy.
Outcomes	Synthetic unvaccinated shares, demand and spillovers.

Hospital labels do not verify cold-chain capability. The county filter excludes facilities outside Nyamira.

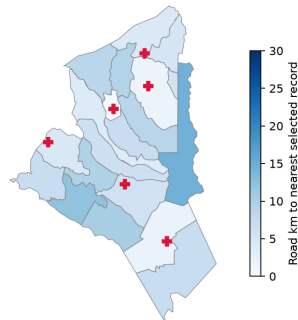
[Sources, licences and provenance](#)

Facility selection changes the accessibility proxy

OSM tagged facilities (6 records)
Mean 9.0 km; maximum 28.2 km



Maina hospital labels (6 records)
Mean 6.8 km; maximum 14.6 km



Boundaries: GADM; roads/facilities: © OpenStreetMap contributors (ODbL); facilities: Maina et al. (2019).

OSM tags and Maina hospital labels differ in coverage, classification and vintage. Neither comparison establishes current vaccine-storage capability.

A bounded, set-based spillover

$$E_i(y) = \min \left(1, \sum_{j \neq i} e^{-\bar{d}_{ij}/R} y_j \right), \quad \mu_i(y) = b_i(1 + \gamma E_i(y)).$$

- Negative γ : cannibalisation. Positive γ : mobilisation.
- Base demand uses population, a simulated unvaccinated share and accessibility.
- The visited set changes turnout; visit order changes only travel.

This simplified mechanism has no chronological patient movement or population-conservation model.

Expected service includes the capacity limit

$$S_i = \min(C, \mu_i \epsilon_i), \quad \epsilon_i \sim \text{LogNormal}(-\sigma^2/2, \sigma).$$

The noise has mean one. With $a = (\log(C/\mu) + \sigma^2/2)/\sigma$,

$$h(\mu) = \mathbb{E}[\min(C, \mu \epsilon)] = \mu \Phi(a - \sigma) + C \Phi(-a).$$

- Train predictors of capped service, using noisy observations.
- Evaluate every route with this exact expectation.
- Optimise the known simulator expectation with a bounded interpolation error.

A deliberately informative synthetic history

log pop	Unvaccinated	Accessibility	Exposure	Served
10.42	0.36	0.48	0.00	237.27
10.79	0.41	0.60	0.00	250.00
10.32	0.49	1.00	0.00	250.00

- 1500 training periods, 400 independent held-out periods.
- Random deployments of zero to ten wards, including zero exposure.
- Potential service at every ward, including unvisited wards.
- These displayed rows come from the actual generated training data.

Real records introduce missing outcomes, capacity censoring and selective historical routing. They need a separate identification and validation strategy.

The same predictor, two ways to plan

Predictor	Static policy	Embedded policy
Linear regression	Exposure fixed at zero	Exposure depends on y
ReLU network	Exposure fixed at zero	Exposure depends on y

- Both policies use the same fitted model and training history.
- Predictions are clipped consistently to $[0, C]$.
- A historical-mean forecast is an additional static baseline.
- Validation uses held-out deployments, not training R^2 .

Routing with decision-dependent service

$$\max \sum_i z_i - \lambda \sum_{i \neq j} d_{ij} x_{ij}, \quad 0 \leq z_i \leq C y_i, \quad z_i \leq \hat{h}_i.$$

- y_i chooses serviced wards; x_{ij} chooses travel arcs.
- An open path starts at Township; at most K wards receive service.
- With embedding, $\hat{h}_i = \max\{0, g(f_i, E_i(y))\}$ changes with the plan.
- The simulator benchmark approximates $h(\mu_i(y))$ from below.

Benchmark upper bound = solver bound + at most 0.5 dose-equivalent interpolation error.
Route values use the exact expectation. Complete models are in the backup slides.

Environment for these results

11th Gen Intel(R) Core(TM) i7-1185G7 @ 3.00GHz

4 cores, 8 logical processors; 63.7 GiB RAM.

Windows-11-10.0.26200-SP0; Python 3.12.14; Gurobi 13.0.3.

scikit-learn 1.9.1; gurobi-machinelearning 1.7.0.

2 solver threads; 60 s limit; relative MIP gap $1e-06$.

Run: 2026-09-29T09:02:56 UTC; full versions, seeds and hashes in results/environment.json.

- Model-building and solver times are recorded separately.
- Cached geography; no imported incumbent or warm start.
- Background load is uncontrolled and unmeasured.
- These are single-machine teaching results, not a runtime benchmark.

Cannibalisation: doses, distance and objective

Policy	Expected doses	km	Objective
static linear	1614.7	78.9	1535.8
embedded linear	1614.7	78.9	1535.8
static MLP	1541.0	73.1	1467.9
embedded MLP	1614.7	78.9	1535.8
historical mean	1614.7	78.9	1535.8
benchmark	1614.7	78.9	1535.8

Benchmark optimum interval: [1535.83, 1536.23]

- Compare static and embedded policies within each predictor.
- More service can require more driving. Doses/km is a different metric.
- The historical baseline is competitive in this synthetic setting.

Synthetic outcomes; objective = expected doses minus one dose-equivalent per km.

Mobilisation: a benefit is not guaranteed

Policy	Expected doses	km	Objective
static linear	1883.0	62.2	1820.8
embedded linear	1883.0	55.7	1827.3
static MLP	1883.6	74.1	1809.5
embedded MLP	1875.4	52.5	1822.9
historical mean	1883.5	55.5	1828.0
benchmark	1880.5	51.8	1828.7

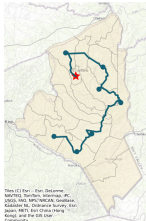
Benchmark optimum interval: [1828.69, 1829.18]

- A fitted model may misrepresent decision-dependent service.
- Embedding does not guarantee a better decision than a careful forecast.
- The benchmark interval quantifies numerical uncertainty.

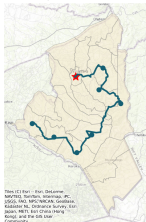
Synthetic outcomes; all policies use the same capacity and road costs.

The selected routes

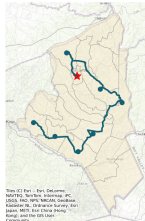
Cannibalisation: static MLP
1541 expected doses; 73.1 km



Mobilisation: static MLP
1884 expected doses; 74.1 km



Cannibalisation: embedded MLP
1615 expected doses; 78.9 km



Mobilisation: embedded MLP
1875 expected doses; 52.5 km



Cannibalisation: benchmark
1615 expected doses; 78.9 km

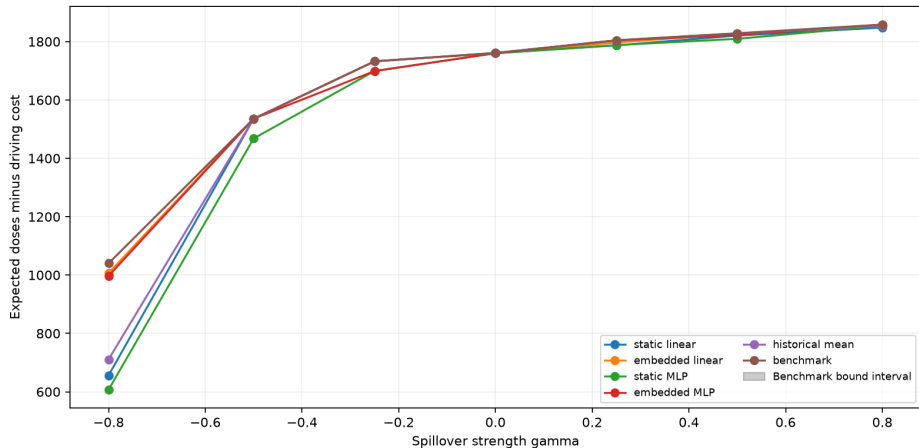


Mobilisation: benchmark
1880 expected doses; 51.8 km



GADM boundaries; roads © OpenStreetMap contributors (ODbL). Synthetic outcomes.

Sensitivity to the assumed interaction



Each strength uses matched histories. The band includes the benchmark's solver and interpolation bounds.

Live demo: a prediction before each run

- ① Change the interaction range from 10 km to 5 or 20 km.
- ② Change the stop budget from 8 to 4 or 12.
- ③ Reduce the network from 8 hidden units to 2, keeping its training data fixed.

Predict expected doses, driving distance and their weighted objective separately.

The notebook also compares three independent training histories. Rerun times depend on the machine and the model; solver limits and gaps stay visible.

[Open the notebook in Colab](#)

What would make this an operational study?

- Actual service and collection locations, verified cold-chain arrangements and travel/time constraints.
- A defensible model of spillover, observed and censored outcomes, and historical deployment decisions.
- Held-out decision performance, geographic validation and programme-specific fairness requirements.

The teaching conclusion

Decisions can change the quantities we predict. Embedding makes that dependence explicit; its value still depends on the model, the data and the objective.

[Code, results, sources and slides](#)

Backup: model notation and inputs

Symbol	Meaning
N, A	Wards; directed arcs $A = \{(i, j) \in N^2 : i \neq j\}$.
s, d_{ij}	Township depot; directed shortest-road distance in km.
K, C, λ	Stop limit (8), capacity per stop (250), cost per km (1).
P_i, u_i, a_i	Population, simulated unvaccinated share, normalised hospital distance.
f_i, b_i	Features $f_i = (\log P_i, u_i, a_i)$; base demand $b_i = .02P_i u_i (.5 + a_i)$.
R, γ, σ	Interaction range (10 km), spillover strength (> -1), noise (.15).

$x_{ij} \in \{0, 1\}$ selects travel arcs; $y_i \in \{0, 1\}$ selects service stops; $z_i \geq 0$ is modelled service. Exposure and predictor outputs are continuous auxiliaries.

The depot receives service and counts toward K . Capacity is per stop; there is no total vehicle-inventory constraint.

Backup: complete routing and service formulation

$$\begin{aligned} \max \quad & \sum_{i \in N} z_i - \lambda \sum_{(i,j) \in A} d_{ij} x_{ij} \\ \text{s.t.} \quad & y_s = 1, \quad \sum_{i \neq s} x_{is} = 0, \quad \sum_i y_i \leq K, \\ & \sum_{j \neq i} x_{ji} = y_i && i \in N \setminus \{s\}, \\ & \sum_{j \neq i} x_{ij} \leq y_i && i \in N, \\ & \sum_{(i,j) \in A} x_{ij} = \sum_i y_i - 1, \\ & \sum_{i,j \in S: i \neq j} x_{ij} \leq \sum_{i \in S} y_i - y_k && \emptyset \neq S \subseteq N \setminus \{s\}, \quad k \in S, \\ & 0 \leq z_i \leq C y_i, \quad z_i \leq \hat{h}_i && i \in N, \\ & x_{ij} \in \{0, 1\}, \quad y_i \in \{0, 1\} && (i,j) \in A, \quad i \in N. \end{aligned}$$

The subtour family is added lazily. These constraints give one open path from s , including the one-stop route. The next slides define $\hat{h}_i$ for every policy.

Backup: exposure and policy-specific service

For embedded policies and the simulator benchmark, for every $i \in N$:

$$w_{ii} = 0, \quad w_{ij} = \exp\left(-\frac{d_{ij} + d_{ji}}{2R}\right), \quad r_i = \sum_j w_{ij} y_j, \quad E_i = \min\{1, r_i\}.$$

Bounds: $0 \leq r_i \leq \max_k \sum_j w_{kj}$ and $0 \leq E_i \leq 1$.

$$\hat{h}_i = \begin{cases} \max\{0, g(f_i, 0)\}, & \text{static linear or MLP,} \\ \max\{0, g(f_i, E_i)\}, & \text{embedded linear or MLP,} \\ T^{-1} \sum_{t=1}^T S_{it}, & \text{historical mean.} \end{cases}$$

All policies also impose $z_i \leq Cy_i$. Maximising z_i therefore clips service to $[0, C]$ at visited wards and gives zero at unvisited wards.

In the static models these service bounds are precomputed constants. The embedded models impose the min/max relations as exact general constraints.

Backup: the fitted predictor inside the model

For input $v_i = (f_i, e_i) \in \mathbb{R}^4$, with $e_i = 0$ (static) or E_i (embedded), apply the fitted scaler:
 $t_{i\ell} = (v_{i\ell} - m_\ell)/s_\ell$, $\ell = 1, \dots, 4$.

Linear regression

$$q_i = \beta_0 + \sum_{\ell=1}^4 \beta_\ell t_{i\ell}, \quad \hat{h}_i = \max\{0, q_i\}.$$

One-hidden-layer ReLU network (default $H = 8$)

$$a_{ih} = c_h + \sum_{\ell=1}^4 B_{h\ell} t_{i\ell}, \quad v_{ih}^+ = \max\{0, a_{ih}\}, \quad h = 1, \dots, H,$$

$$q_i = c_0 + \sum_{h=1}^H \alpha_h v_{ih}^+, \quad \hat{h}_i = \max\{0, q_i\}.$$

All scaler and predictor coefficients are fixed after training. Affine auxiliaries are real; ReLU and clipped outputs are nonnegative. The min/max constraints have exact mixed-integer representations in Gurobi.

Backup: synthetic service and its expectation

For each ward and deployment, the teaching simulator specifies

$$\mu_i = b_i(1 + \gamma E_i), \quad \epsilon_i \sim \text{LogNormal}(-\sigma^2/2, \sigma), \quad S_i = \min\{C, \mu_i \epsilon_i\}.$$

For $\mu > 0$ and $\sigma > 0$, with $a = (\log(C/\mu) + \sigma^2/2)/\sigma$,

$$h(\mu) = \mathbb{E}[\min\{C, \mu \epsilon\}] = \mu \Phi(a - \sigma) + C \Phi(-a).$$

Here Φ is the standard normal CDF. The limiting cases are $h(0) = 0$ and, for $\sigma = 0$, $h(\mu) = \min\{C, \mu\}$.

The predictors are fitted to noisy capped outcomes S_{it} from 1500 simulated periods; 400 independent periods supply validation. Every ward has a potential outcome in every period. Model fitting precedes routing; the routing solver does not retrain the predictor.

Backup: simulator benchmark and route evaluation

Write $H_i(e) = h(b_i(1 + \gamma e))$, using the expected capped service function. On each interval $[e_{i\ell}, e_{i,\ell+1}]$ of a grid covering $[0, 1]$, set

$$a_{i\ell} = \frac{H_i(e_{i,\ell+1}) - H_i(e_{i\ell})}{e_{i,\ell+1} - e_{i\ell}}, \quad c_{i\ell} = H_i(e_{i\ell}) - a_{i\ell}e_{i\ell}.$$

Keep all routing, exposure and capacity constraints. Replace the predictor bound by

$$z_i \leq a_{i\ell}E_i + c_{i\ell} \quad \text{for every } i, \ell.$$

Concavity makes the minimum of these lines a secant under-estimator of H_i . When $\sigma = 0$, use $z_i \leq b_i(1 + \gamma E_i)$ together with $z_i \leq Cy_i$.

Evaluate every returned route (x^*, y^*) with the same exact expected objective:

$$F(x^*, y^*) = \sum_i y_i^* h(b_i(1 + \gamma E_i(y^*))) - \lambda \sum_{(i,j) \in A} d_{ij} x_{ij}^*.$$

Backup: a bound on benchmark interpolation

Expected capped service is concave in exposure. For $\sigma > 0$,

$$\left| \frac{d^2 h(b_i(1 + \gamma E))}{dE^2} \right| \leq M_i = \frac{(b_i \gamma)^2 e^{\sigma^2}}{C \sigma \sqrt{2\pi}}.$$

A secant over width Δ underestimates by at most $M_i \Delta^2/8$. Choose each ward's grid so its error is at most ε/K .

$$\text{exact feasible route value} \leq \text{optimal expected value} \leq \text{solver bound} + \varepsilon.$$

Here $\varepsilon = 0.5$ dose-equivalent. Tests check the expectation against numerical integration and small routing instances against exhaustive search.